AN UNSUPERVISED DEEP LEARNING METHOD FOR SUBSURFACE TARGET DETECTION IN RADAR SOUNDER DATA

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ABSTRACT

Radar sounder data are widely used for investigating geological structures and processes in the subsurface of icy and arid areas. Visual interpretation is one of the main techniques used in the literature to extract information from radargrams. There exist some automatic approaches but mostly supervised. However, no methods exploit deep learning in an unsupervised way. Here, we propose an automatic and unsupervised technique for extracting information on the subsurface geological targets. The technique is built upon three steps: i) generation of a coarse segmentation map based on the radargram statistical properties, ii) refinement of the coarse map with deep learning to detect target reflections, and iii) analysis of the deep features to identify buried targets. We tested the proposed method on MARSIS radar data acquired near the South Pole of Mars. The experimental results prove the effectiveness of the proposed method.

Index Terms— deep learning, statistical analysis, subsurface target detection, radar sounder

1. INTRODUCTION

Radar sounders (RSs) are active sensors that are widely used to probe the subsurface of icy and arid areas. RSs transmit an electromagnetic (EM) wave at the nadir with a low frequency in the range of high or very high frequencies (HF-VHF) and with a wide bandwidth. Such EM waves can penetrate the surface and propagate into the subsurface. In the propagation, the wave interacts with the dielectric interfaces that backscatter part of the signal to the antenna. All backscattered echoes are represented in radargrams. Interfaces are related to the dielectric contrast between different materials. Hence, radargrams capture the buried geological structures.

Radargrams are of extreme importance for a better understanding of geological targets and processes in the subsurface. During the past decades, visual interpretation of radargrams unveiled a large amount of information on the subsurface geological structures and processes in icy and arid areas on the Earth, Mars, and the Moon. Radargrams of the cryosphere mainly represent the reflections from the surface and ice layers, the basal return (aka basal return), the noise above the surface and below the basal return, and the echo-free zone (EFZ). Recently, visual inspection of Mars radargrams highlighted subsurface anomalies at the Martian South Pole basal return that can be interpreted as subglacial lakes [1]. However, considering the increasing amount of radargrams available from satellite and airborne missions and the planned RS acquisitions, it is mandatory to develop accurate automatic techniques that can support the analysis of RS data in a timely and reliable manner. In the literature, few automatic methods have been proposed for the automatic extraction of information from radargrams. They are mostly based on a supervised approach [2]. Recently, few deep learning methods have been proposed for analyzing radar sounder data that are also based on supervised methods for segmenting inland radargrams and detecting internal layers [3]. Despite their intrinsic effectiveness, deep learning-based approaches require a large amount of labeled data for network training. Unsupervised algorithms are needed, considering the difficulties in retrieving reliable RS-labeled datasets. Existing methods mostly rely on classic machine learning and statistical analysis [4, 5], which in many application domains have been overcome by deep learning models.

This paper proposes a novel and unsupervised method based on deep learning for extracting information from RS data. The proposed approach aims at detecting subsurface reflections in radargrams by analyzing the related deep features to identify buried targets. The proposed method consists of three main steps– i) generation of a coarse segmentation map based on the statistical properties of the targets, ii) refinement of the coarse map with deep learning to detect target reflections, and iii) analysis of the deep features of the target reflections to identify buried targets. We tested the proposed method on a dataset of MARSIS radargrams acquired near the South Pole of Mars to identify anomalies in the basal return that astrophysicists identified as subglacial lakes [1].

2. PROPOSED DEEP LEARNING METHOD

Let us consider a radargram as a 2D matrix of N_T traces and N_S samples defined as $\mathbf{R} = \{R(a,r)|a \in [1,...,N_T], r \in [1,...,N_S]\}$, where R is the power of the reflections in the radargrams, and (a,r) the azimuth and the range coordinates,

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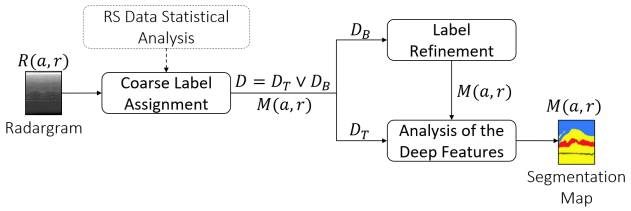


Fig. 1: Block scheme of the proposed deep approach.

respectively. Each radar acquisition corresponds to a trace of the radargram. Thus, adjacent pixels capture the same geologic targets in a given area of the subsurface. This paper proposes a novel unsupervised method for detecting subsurface targets in radargrams. The technique is fully unsupervised and does not require any initial labeled data.

We model radargrams with two main semantic classes: the background and the target. The background includes the no-signal areas and the thermal noise above the surface and below the basal return. The targets include the reflections from different dielectric interfaces in the subsurface, such as ice layers and basal return. Thus, the target class can be further divided into other subclasses. This hierarchical division is exploited by the proposed method to segments the radargram. At first, we discriminate between the background and target to extract reflections in the radargrams. Then, analyzing the deep features, the target (i.e., radargram reflections) is segmented into subclasses $\omega_c, c \in [1, \ldots, N_\Omega]$. The method consists of three main steps (see the flow chart in Fig. 1)– i) generation of a coarse segmentation map $M(a, r)$ by exploiting the statistical properties of RS data, ii) refine the coarse segmentation map $M(a, r)$ to detect target reflections with deep learning, and iii) hierarchical update of $M(a, r)$ by analyzing deep features of the target reflections.

2.1. Candidate Label Assignment

Considering the radargram statistical properties, the first step generates a candidate label for all the pixels that will be refined using a deep network. Here, we plan to use a convolutional neural network (CNN) architecture, as this model showed good performance in analyzing 2D data. As CNNs require patches in the input, we divide the input radargrams into patches of dimensions (w_p, h_p) to compose the dataset $\mathcal{D}$. $\mathcal{D}$ is further divided into two subsets: $\mathcal{D}_\mathcal{B}$ and $\mathcal{D}_\mathcal{T}$ so that $\mathcal{D} = \mathcal{D}_\mathcal{B} \vee \mathcal{D}_\mathcal{T}$. $\mathcal{D}_\mathcal{B}$ contains the patches only imaging the background, and thus, the candidate label of the pixels in these patches is that of the background ω_B. $\mathcal{D}_\mathcal{T}$ contains patches mostly capturing targets, and thus, the candidate label of the pixels in these patches is that of the target ω_T. Note that patches in $\mathcal{D}_\mathcal{B}$ contain only background pixels, while patches in $\mathcal{D}_\mathcal{T}$ image background and target pixels generated by different geological interfaces. The background patches are statistically more homogeneous than the target patches as they capture just one class. The target patches are less homogeneous as they image the background and reflections from different targets.

To separate target and background patches, we exploited the statistical properties of the two classes. Considering the method in [5], we fitted several distributions (i.e., Rayleigh, Nakagami, Gamma, K, and Weibull) to radargram targets to find the best fitting pdf. The best-fitting distribution p_{best} is the one minimizing the Mean Square Error (MSE) and the Kullback-Leibler (KL) divergence. The parameter defining p_{best} depends on the best fitting distribution– if the best distribution is the K pdf, $p_{best} = p_K$ and the parameters are $\theta_{best} = (\mu_z, \beta_K)$. If the best distribution is the Gamma pdf, $p_{best} = p_\Gamma$ and the best parameters are $\theta_{best} = (\alpha_\Gamma, \beta_\Gamma)$, and so on for the other distributions. The patches in $\mathcal{D}$ are fitted with the p_{best} distribution to estimate the parameter $\tilde{\theta}_{best}$. Hence, a vector $\tilde{\theta}$ is generated that contains for each patch P_p the estimated parameters $\tilde{\theta}_p$. The parameter $\tilde{\theta}_p$ is representative of the class imaged by the p-th patch. The histogram of $\tilde{\theta}$ shows a bimodal distribution: one for the target patches, and the other for the background patches. The two modes can be separated with a threshold T_θ. Here, we used Otsu's method [6] as it has good performance with histograms with a bimodal distribution, but any other approach in the literature can be used. For each patch P_p, $\tilde{\theta}_p$ is thresholded:

$$P_p \in \begin{cases} D_B & \text{if } \tilde{\theta}_p \leq T_\theta \\ D_T & \text{if } \tilde{\theta}_p > T_\theta \end{cases} \quad (1)$$

Finally, we define the segmentation map $M(a, r)$, where pixels in the patches in D_B are labeled with ω_B, and pixels in the patches in D_T are labeled with ω_T.

2.2. Label Refinement for Target Reflection Detection

This step updates the labels in $M(a, r)$ to extract reflections generated by dielectric interfaces in the subsurface. To this end, we updated $M(a, r)$ to generate a binary map for each pixel (a, r) to distinguish background and target pixels. We train a neural network in an unsupervised manner with the task of reconstructing patches in $\mathcal{D}_B$ (background patches). Here, we use a W-Net [7] as it showed good performances in many applications. Each convolutional layer learns semantic features from the spatial and semantic viewpoint related to the background and not the target. Testing such a network with $\mathcal{D}$, we expect to have an excellent reconstruction of background patches and a poor reconstruction of target patches. Similar to [8], we can exploit the normalized reconstruction error map $\mathcal{E}(a, r)$ defined for all the pixels (a, r) to identify pixels representing the background and the targets. Background pixels have a small reconstruction error, while target pixels have a larger reconstruction error. We update the pixels labeled as ω_T in $M(a, r)$ so that pixels with small error belong to the background (ω_B), and pixels with large error

belong to the target class (ω_T). Reflections can be identified as groups of pixels similarly labeled that are not separated by background pixels.

2.3. Analysis of Deep features

The third step aims at analyzing the pixels labeled as ω_T in $M(a,r)$ to identify the subclass labels $\omega_c, c \in [1,\dots,N_c]$, i.e., discriminate between different geological targets, by constructing a decision tree considering feature representative of each class. As the network is trained only with background patches and we want to analyze subclasses of the target class, we train the network with $\mathcal{D}_\mathcal{T}$ with the task of reconstructing the related patches. Then, we extract deep features for the patches in $\mathcal{D}_\mathcal{T}$ from N_L convolutional layers, and we upsampled them to the original patch size (h_p, w_p) with bilinear interpolation. The deep features are masked by $M(a,r)$ to keep only those for the target pixels labeled with ω_T, obtaining $\mathcal{F}(a,r,k)$.

$\mathcal{F}$ is analyzed with an approach similar to [9] for hierarchically labeling target pixels with ω_c, $c \in [1,\dots,N_\omega]$ by constructing a decision tree. The decision tree is built by selecting semantically meaningful features. The number of features extracted from the deep network is large, which increases the complexity of the problem. In the W-Net, the maximum feature number per layer is 512 and the minimum 64. The features of the different layers are min-max normalized and then concatenated together to obtain $\mathcal{G} = g^k(a,r), k \in [1,...,N_G]$.

We discretize the features to reduce the problem complexity and the feature sparsity while preserving the information. We binarize $\mathcal{G}$ so that features with g^k greater than 0.5 are set to 1 and those with g^k smaller than 0.5 to 0. Each class is expected to correspond to a specific binary signature. To select the most informative feature k^* to build the decision tree, we followed the approach in [9]. Hence, features are analyzed by considering the index R_k based on the Hamming distance (number of different pixels). k^* is defined as the one that maximizes R_k, and it is used to divide the pixels into two classes to build the decision tree and update $M(a,r)$. Features similar to k^*, i.e., those having a small Hamming distance, are discarded. This approach is iterated to divide target pixels and update $M(a,r)$.

3. DATASET AND EXPERIMENTAL RESULTS

To prove the proposed method's effectiveness, we analyzed a dataset of radargrams acquired by the MARSIS radar sounder near the South Pole of Mars. MARSIS transmits chirp pulses with 1 MHz bandwidth and centered at 1.8, 3, 4, or 5 MHz, alternating two frequencies. The dataset consists of 37 radargrams with High and Medium SNR in [1]. Each radargram has $N_T = 3200$ traces and $N_S = 980$ samples and covers a track of 100 km long. We aim at discriminating the

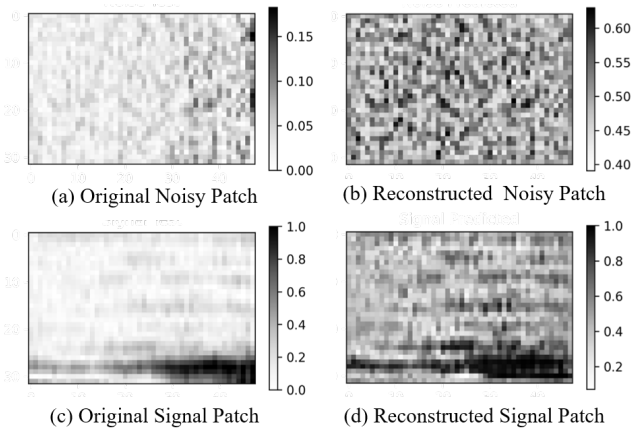


Fig. 2: Two examples of pairs of input and reconstructed patches: (a) and (b) background, and (c) and (d) target.

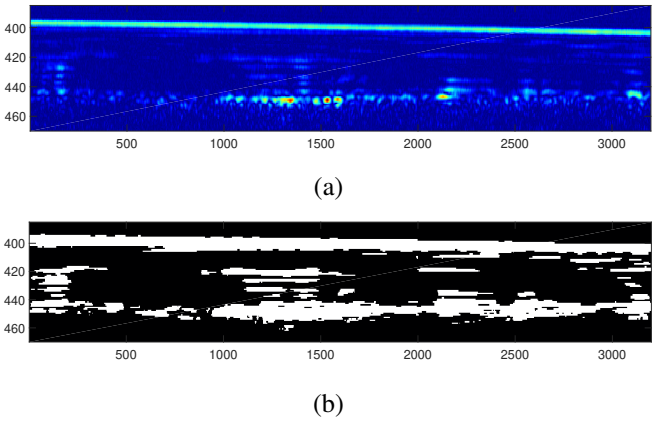


Fig. 3: (a) Input radargram 10737 acquired at 4MHz, (b) $M(a,r)$ at the end of the label refinement step (binary map).

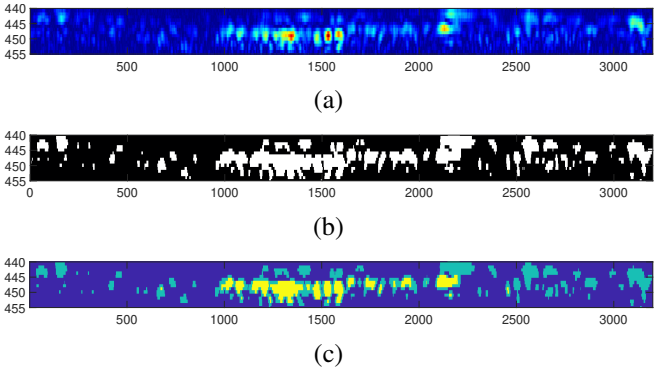


Fig. 4: Zoom of the basal return of (a) radargram 10737 acquired at 4MHz, (b) $M(a,r)$ at the end of the label refinement step, and (c) segmented map (blue indicate the background, green indicates non-anomaly and yellow the anomalies).

Table 1: Table with the performance of the classes of the basal return segmentation over the whole dataset.

Class	Accuracy	Sensitivity	Specificity
Background	88.87%	0.8890	0.8681
Non-Anomalies	78.45%	0.9025	0.8937
Anomalies	76.54%	0.8956	0.8715

characteristics of the basal return between non-anomalies or anomalies, which can be interpreted as rocky interface and subglacial lakes, respectively [1].

Hence, the method has to detect reflections in the radargram (i.e., the ice layers and the basal return) and then discriminate the basal return pixels between non-anomalies (ω_1) and anomalies (ω_2). The results in [1], i.e., the subglacial lakes detected by analyzing the statistics of the bedrock, are considered as a reference to qualitatively and quantitatively evaluate the proposed method. The radargrams are preprocessed with range compression and geometric calibration to compensate for altitude variations. Since the surface of the investigated area is flat and the roughness is low at the MARSIS wavelength, the scattering is mostly coherent and clutter can be neglected. SAR focusing is not required as it would reduce to a moving average filtering [1].

For generating the dataset, we consider patches of dimensions $w_p = 48$ and $h_p = 32$ as it is small enough to capture one class and large enough to be processed by the network. We split the 37 radargrams into 75480 patches: 59160 patches are used for the training and 16320 for the test. The two datasets are independent as the patches for the training and test are extracted from different radargrams. The best-fitting distribution is the Nakagami pdf. For dividing target and background patches, the threshold is $T_\theta = 1.6$. The W-Net is trained with a learning rate of 10^{-4} and 100 epochs. Fig. 2 shows two examples of pairs of input and reconstructed patches of the background (Fig.2.a and b) and target (Fig.2.c and d). Fig.3 shows the input radargram and the binary map $\mathcal{M}(a, r)$ at the end of the label refinement step for the basal return for the orbit 10737 acquired at 4MHz. The performance of the label refinement step over the dataset of 37 radargrams is good. The overall accuracy is equal to 88.87%, and the sensitivity to 0.889, and the specificity to 0.8681 (see Table 1). Fig.4 shows a zoom over the basal return of (a) the input radargram, (b) the binary map $M(a, r)$ at the end of the label refinement step, and (c) the segmented map. The background is in blue, the non-anomaly of the basal return is in green, and the anomalies are yellow. Table 1 shows the proposed method performance for the segmentation map for the basal return averaged for the 37 radargrams. The background class accuracy is 88.87%, that of the non-anomaly is equal to 78.45%, and that of the anomalies to 76.54%. The performance proves the effectiveness of the method in detecting the reflections and identifying targets of interest.

4. CONCLUSIONS

This paper has proposed a deep learning-based method for analyzing radargrams in an unsupervised and automatic way. The proposed method generates a binary map from which it is possible to detect the reflections from subsurface interfaces. Then it analyses deep features extracted by a W-Net to discriminate the subsurface targets in an unsupervised way. The proposed method was validated by detecting layer reflections in planetary radargrams and then identifying the basal return and its characteristics, i.e., rocky interface or anomalies. The radargrams were acquired by the MARSIS radar sounder in the South Pole of Mars. For the experimental analysis, we used an NVIDIA™ Tesla T4 GPU with 15 GB of RAM. Both training phases took about 7.5 hours, and the testing, feature extraction, and analysis took about one hour. The results show the effectiveness of the proposed method. As future work, we plan to perform an extensive analysis considering also other target classes on data acquired in other scenarios, such as the Earth, and by other sounders with different parameters.

5. REFERENCES

[1] R Orosei et al., "Radar evidence of subglacial liquid water on mars," *Science*, vol. 361, no. 6401, pp. 490–493, 2018.

[2] E Donini et al., "An automatic approach to map refreezing ice in radar sounder data," in *Image and Signal Processing for Remote Sensing XXV*. International Society for Optics and Photonics, 2019, vol. 11155, p. 111551B.

[3] M Rahnemoonfar et al., "Deep multi-scale learning for automatic tracking of internal layers of ice in radar data," *Journal of Glaciology*, pp. 1–10, 2020.

[4] E Donini et al., "An unsupervised fuzzy system for the automatic detection of candidate lava tubes in radar sounder data," *IEEE Transactions on Geoscience and Remote Sensing*, pp. 1–19, 2021.

[5] A Ferro and L Bruzzone, "Analysis of radar sounder signals for the automatic detection and characterization of subsurface features," *IEEE Transactions on Geoscience and Remote Sensing*, vol. 50, no. 11, pp. 4333–4348, 2012.

[6] H Otsu, "A threshold selection method from gray-level histograms," *IEEE transactions on systems, man, and cybernetics*, vol. 9, no. 1, pp. 62–66, 1979.

[7] X Xia and B Kulis, "W-net: A deep model for fully unsupervised image segmentation," *arXiv preprint arXiv:1711.08506*, 2017.

[8] LE Besaw and PJ Stimac, "Deep learning algorithms for detecting explosive hazards in ground penetrating radar data," in *Detection and Sensing of Mines, Explosive Objects, and Obscured Targets XIX*. International Society for Optics and Photonics, 2014, vol. 9072, p. 90720Y.

[9] S Saha et al., "Unsupervised deep change vector analysis for multiple-change detection in vhr images," *IEEE Transactions on Geoscience and Remote Sensing*, vol. 57, no. 6, pp. 3677–3693, 2019.